

应用数学导论第二次作业

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1. 称弹性体中的位移场 $\mathbf{u} = (u, v, w)^T$ 为刚体位移场, 若其对应的应变张量 $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ 恒为 $\mathbf{O}$. 试显式给出刚体位移场的一般形式.

解答.

由于

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T) = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}) \\ \frac{1}{2}(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}) & \frac{\partial v}{\partial y} & \frac{1}{2}(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}) \\ \frac{1}{2}(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}) & \frac{1}{2}(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}) & \frac{\partial w}{\partial z} \end{pmatrix} = \mathbf{O},$$

故

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial w}{\partial z} = 0, \quad \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = 0,$$

从而有

$$\frac{\partial^2 u}{\partial y \partial z} + \frac{\partial^2 v}{\partial z \partial x} = \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) = 0,$$

同理可得

$$\begin{cases} \frac{\partial^2 u}{\partial y \partial z} + \frac{\partial^2 v}{\partial z \partial x} = 0, \\ \frac{\partial^2 v}{\partial z \partial x} + \frac{\partial^2 w}{\partial x \partial y} = 0, \\ \frac{\partial^2 w}{\partial x \partial y} + \frac{\partial^2 u}{\partial y \partial z} = 0, \end{cases}$$

解得

$$\frac{\partial^2 u}{\partial y \partial z} = \frac{\partial^2 v}{\partial z \partial x} = \frac{\partial^2 w}{\partial x \partial y} = 0.$$

由于

$$\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial y \partial z} = 0,$$

故设 $u = f(y, z)$, 其中

$$\frac{\partial}{\partial z} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 u}{\partial y \partial z} = 0, \quad \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial z} \right) = \frac{\partial^2 u}{\partial y \partial z} = 0,$$

从而可设

$$\frac{\partial f}{\partial y} = f_1(y), \quad \frac{\partial f}{\partial z} = f_2(z),$$

于是 $u = f(y, z)$ 一定可以表示为 $u = u_1(y) + u_2(z)$, 同理可设 $v = v_1(z) + v_2(x)$, $w = w_1(x) + w_2(y)$, 其中一元函数 $u_1, u_2, v_1, v_2, w_1, w_2$ 须满足:

$$\frac{du_1}{dy} + \frac{dv_2}{dx} = \frac{du_2}{dz} + \frac{dw_1}{dx} = \frac{dv_1}{dz} + \frac{dw_2}{dy} = 0.$$

注意到 $\frac{du_1}{dy}$ 是关于 y 的函数, 而 $\frac{dv_2}{dx}$ 是关于 x 的函数, 因此只能有

$$\frac{du_1}{dy} = a_1, \quad \frac{dv_2}{dx} = -a_1,$$

故 $u_1(y) = a_1 y + m_1$, $v_2(x) = -a_1 x + m_2$, 同理可得其他 2 组式子, 因此最后得到刚体位移场的一般形式为

$$\mathbf{u} = \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} a_1 y - a_3 z \\ a_2 z - a_1 x \\ a_3 x - a_2 y \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \quad a_1, a_2, a_3, b_1, b_2, b_3 \in \mathbb{R},$$

还可表示为

$$\mathbf{u} = \mathbf{a} \times \mathbf{x} + \mathbf{b}, \quad \mathbf{a} := \begin{pmatrix} -a_2 \\ -a_3 \\ -a_1 \end{pmatrix}, \quad \mathbf{x} := \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} := \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix},$$

由此得到刚体的位移可分解为转动和平移. □

2. 试证明变形协调方程的第一式和第四式:

$$\frac{\partial^2 \varepsilon_y}{\partial x^2} + \frac{\partial^2 \varepsilon_x}{\partial y^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y},$$

$$\frac{\partial}{\partial x} \left(-\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{zx}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z} \right) = 2 \frac{\partial^2 \varepsilon_x}{\partial y \partial z}.$$

解答.

直接代入

$$\varepsilon_x = \frac{\partial u}{\partial x}, \quad \varepsilon_y = \frac{\partial v}{\partial y}, \quad \varepsilon_z = \frac{\partial w}{\partial z}, \quad \gamma_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}, \quad \gamma_{yz} = \frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}, \quad \gamma_{zx} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x},$$

计算可得

$$\begin{aligned}\frac{\partial^2 \varepsilon_y}{\partial x^2} + \frac{\partial^2 \varepsilon_x}{\partial y^2} &= \frac{\partial^3 v}{\partial x^2 \partial y} + \frac{\partial^3 u}{\partial x \partial y^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y}, \\ \frac{\partial}{\partial x} \left(-\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{zx}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z} \right) &= 2 \frac{\partial^3 u}{\partial x \partial y \partial z} = 2 \frac{\partial^2 \varepsilon_x}{\partial y \partial z}.\end{aligned}$$

□